

Program - 4

Project Structure :

program4/

- Dockerfile
- package.json
- package-lock.json
- index.js
- node_modules

Step 1: Create the Dockerfile

Step 2: create a file -> index.js

Step 3: Create a file -> package.json by doing npm init -y

Step 4: Install Node modules with the following command npm install express

```
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab$ mkdir Program4
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab$ cd Program4
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab/Program4$ touch index.js
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab/Program4$ touch Dockerfile
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab/Program4$ open index.js
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab/Program4$ open Dockerfile
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab/Program4$ npm init -y
Wrote to /home/1rv24mc031_darshan_hegde/Desktop/DevLab/Program4/package.json:
```

```
{
  "name": "program4",
  "version": "1.0.0",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "description": ""
}
```

```
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab/Program4$ npm i express
```

```
added 68 packages, and audited 69 packages in 3s
```

```

1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab/Program4$ cat index.js
const express=require('express')
const app=express()
const PORT=3000

app.get("/",(req,res)=>{
res.send("Hello from git+Docker integration");
});

app.listen(PORT,()=>{
console.log(`server running at http://localhost:${PORT}`);
});
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab/Program4$ cat Dockerfile
#Stage 1 - Build Stage
FROM node:18 AS build
WORKDIR /app
COPY package*.json ./
RUN npm install
COPY . .

# Stage 2 - Runtime Stage
FROM node:18-slim
WORKDIR /app
COPY --from=build /app /app
EXPOSE 3000
CMD ["node","index.js"]

```

How to Generate a Personal Access Token (Classic)

Log in to your GitHub account.

Go to Settings → Developer settings → Personal access tokens → Tokens (classic)

Click Generate new token → Generate new token (classic)

Step 1: Initialize a Local Repository

git init

Rename the default branch to main (recommended):

git branch -M main

Step 2: Add and Commit Files

git add .

Commit them with a message:

git commit -m "Initial commit"

```

1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab/Program4$ git init
Initialized empty Git repository in /home/1rv24mc031_darshan_hegde/Desktop/DevLab/Program4/.git/
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab/Program4$ git add .
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab/Program4$ git commit -m "First commit"
[main (root-commit) 9710f31] First commit
 636 files changed, 73045 insertions(+)
 create mode 100644 Dockerfile
 create mode 100644 index.js
 create mode 100644 node_modules/.package-lock.json
 create mode 100644 node_modules/accepts/HISTORY.md

```

Step 3: Connect to a Remote GitHub Repository

git remote add origin <https://github.com/><your-username>/<repo-name>.git

Step 4: Using username and PAT

git push -u origin main

```

create mode 100644 package.json
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab/Program4$ git remote add origin https://github.com/DarshanHegdeP/Devops_Lab_Program4.git
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab/Program4$ git push -u origin main
Username for 'https://github.com': DarshanHegdeP
Password for 'https://DarshanHegdeP@github.com':
Enumerating objects: 722, done.
Counting objects: 100% (722/722), done.
Delta compression using up to 4 threads
Compressing objects: 100% (667/667), done.
Writing objects: 100% (722/722), 732.32 KiB | 6.60 MiB/s, done.
Total 722 (delta 144), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (144/144), done.
To https://github.com/DarshanHegdeP/Devops_Lab_Program4.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

```

